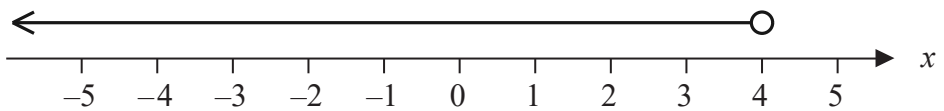
19 (a) On the number line, show the inequality $x < 4$



2 marks total
Open circle at 4 and arrow to the left
4 is excluded; all smaller values are included

(2)

$3 < y \leqslant 7$ where y is an integer.

(b) Write down all the possible values of y.

Greater than 3, and up to and including 7.

2 marks total
All four integers, with no extra values

4, 5, 6, 7

(2)

(c) Solve $3x + 5 \geqslant x + 17$

$3x + 5 \geq x + 17$
$-x \qquad -x$
$2x + 5 \geq 17$
$-5 \qquad -5$
$2x \geq 12$
$\div 2 \qquad \div 2$
$x \geq 6$

1 mark
Subtract x from both sides

1 mark
Subtract 5, then divide by 2

1 mark
Correct inequality

Dividing by positive 2 keeps the inequality sign.

$x \geq 6$

(3)

(Total for Question 19 is 7 marks)